# Alex (Wei) Chen

Director

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## Summary

Director at Qualcomm (via acquisition of Nexa AI). Researcher and engineer focused on AI systems, generative AI inference, hardware-software co-design, and robotics. PhD from Stanford University (2024).

## Experience

**Qualcomm**, Director (Principal Engineer/Manager) — Santa Clara, CA
2026 – present
1 year

- Joined Qualcomm following the acquisition of Nexa AI. Lead generative AI inference research and engineering focused on accelerating LLMs and VLMs on Qualcomm Snapdragon NPU (Hexagon HTP) and Snapdragon SoCs.
- Drive end-to-end optimization of LLMs and VLMs for Qualcomm Hexagon HTP NPU, encompassing hardware-aware quantization (INT4/INT8/FP16 mixed-precision), operator fusion, memory bandwidth scheduling, and kernel-level tuning.
- Collaborate with silicon, compiler, and runtime teams to co-design neural network architectures and inference pipelines that fully exploit Hexagon NPU vector and tensor acceleration.
- Manage a cross-functional team spanning model research, runtime optimization, and SDK development.

**Nexa AI**, Founder, CEO and Chief Scientist — Palo Alto, CA
2024 – 2026
2 years

- Founded Nexa AI, specializing in efficient AI research and deployment. Built the NexaML engine and NexaSDK (7.6K GitHub stars) for generative AI model deployment on NPU, GPU, and CPU. Acquired by Qualcomm in 2026.
- Principal architect and first author of AutoNeural, Octopus V1–V4, OmniVLM, OmniAudio, and NexaQuant. Octopus V2 represents ~2% of total HuggingFace downloads since 2022.
- Solutions deliver ~55% faster throughput and ~28% better output quality vs. existing AI inference stacks. Trusted by Geely, HP, Lenovo, and İşbank.
- Official partner of Qualcomm, AMD, NVIDIA, IBM, Google, Microsoft, Intel, Docker, HP, Lenovo, and Dell.

## Education

**Stanford University** — California, US
2019 – 2024

## Awards

**Best Paper Runner-Up Award** — Jan 2025
For the paper: DP-FedLoRA: Privacy-Enhanced Federated Fine-Tuning for On-Device Large Language Models.
IEEE International Conference on Data Mining (ICDM 2025)

## Publications

**Octopus: On-device language model for function calling of software APIs**
Wei Chen, Zhiyuan Li, Mingyuan Ma
aclanthology.org/2025.naacl-industry.27

**Octo-planner: On-device Language Model for Planner-Action Agents**
Wei Chen, Zhiyuan Li, Zhen Guo, Yikang Shen
arxiv.org/abs/2406.18082

**DP-FedLoRA: Privacy-Enhanced Federated Fine-Tuning for On-Device LLMs**
Honghui Xu, Shiva Shrestha, Wei Chen, Zhiyuan Li, Zhipeng Cai
arxiv.org/abs/2509.09097

**AutoNeural: Co-Designing Vision-Language Models for NPU Inference**
Wei Chen, Liangmin Wu, Yunhai Hu, Zhiyuan Li, et al.
arxiv.org/abs/2512.02924

**OmniVLM: A Token-Compressed, Sub-Billion-Parameter Vision-Language Model for Efficient On-Device Inference**
Wei Chen, Zhiyuan Li, Shuo Xin
arxiv.org/abs/2412.11475

**Dolphin: Long Context as a New Modality for Energy-Efficient On-Device Language Models**
Wei Chen, Zhiyuan Li, Shuo Xin, Yihao Wang
arxiv.org/abs/2408.15518

**Octopus v4: Graph of language models**
Wei Chen, Zhiyuan Li
arxiv.org/abs/2404.19296

**Octopus v3: Technical Report for On-device Sub-billion Multimodal AI Agent**
Wei Chen, Zhiyuan Li
arxiv.org/abs/2404.11459

**Octopus v2: On-device language model for super agent**
Wei Chen, Zhiyuan Li
arxiv.org/abs/2404.01744

**On-Device Language Models: A Comprehensive Review**
Jiajun Xu, Zhiyuan Li, Wei Chen, Qun Wang, Xin Gao, Qi Cai, Ziyuan Ling
arxiv.org/abs/2409.00088

## Skills

**Programming Languages**

**Machine Learning**

**Hardware Acceleration**

**Systems**

## Languages

**English**
Fluent

**Chinese (Mandarin)**
Native